

ESM2-small: A Compact Protein Language Model from Swiss-Prot

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Abstract

We present ESM2-small, a compact 9.6M-parameter Transformer protein language model trained on Swiss-Prot curated protein sequences. The model uses a 12-layer masked language modeling architecture with pre-norm, GELU activation, and 15% masking ratio, trained for 5 epochs on 433,583 sequences. ESM2-small achieves a final validation perplexity of 1.54 and loss of 0.417, comparable to larger models on per-token metrics. Zero-shot evaluation on GFP fluorescence mutations yields a Spearman correlation of 0.200, demonstrating learned evolutionary and functional signals. The model is available at <https://github.com/juniorlp/ESM2-small> with full training code, data pipeline, and pre-trained weights.

Keywords: protein language model, masked language modeling, protein engineering, zero-shot prediction, Swiss-Prot

1 Introduction

Protein language models (PLMs) have emerged as powerful tools for learning representations of protein sequences from evolutionarily-related data [7]. Large-scale PLMs such as ESM-2 [9] (up to 15B parameters) and ProGen2 [13] have demonstrated remarkable zero-shot and few-shot capabilities across diverse downstream tasks including structure prediction, function annotation, and variant effect prediction.

Despite their effectiveness, the computational cost of training and deploying large PLMs remains a significant barrier for many researchers. This has motivated interest in smaller, more efficient models that retain most of the predictive capability [4, 8]. In this work, we address this gap by training ESM2-small, a minimal instantiation of the ESM-2 architecture on Swiss-Prot, and evaluating its zero-shot mutation prediction capability on the GFP benchmark.

Our contributions:

- Training of a 9.6M-parameter ESM-2 architecture on Swiss-Prot curated sequences (433,583 training, 22,821 validation)
- Demonstration of zero-shot fitness prediction on GFP fluorescence mutations with Spearman $\rho = 0.200$
- Public release of full training code, data pipeline, and pre-trained weights

2 Related Work

2.1 Protein Language Models

ESM-2 [9] established that scaling Transformer language models on protein sequences leads to emergent structural and functional understanding. Following this, numerous works have explored training efficiency [4], specialized architectures [11], and domain adaptation [6].

2.2 Zero-Shot Variant Effect Prediction

ESM-1v demonstrated that large PLMs trained with masked language modeling (MLM) can predict the functional effects of protein mutations without task-specific fine-tuning [12]. This zero-shot approach was later extended by ProteinGym [1] and PoET [13], establishing benchmarks for evaluating PLM zero-shot capabilities.

3 Methods

3.1 Model Architecture

ESM2-small mirrors the ESM-2 architecture [9]. The key hyperparameters are summarized in Table 1.

Table 1: ESM2-small Architecture

Parameter	Value
Layers	12
Hidden dimension (d_{model})	256
Attention heads	8
FFN dimension	1024
Vocabulary size	31 (amino acids + special tokens)
Max sequence length	512
Total parameters	9,624,607

The model uses pre-norm residual connections [2], GELU activation [5], and rotary positional embeddings (RoPE) omitted in this ablation-scale model. The MLM objective masks 15% of tokens uniformly at random.

3.2 Training Data

We train on Swiss-Prot [3], a curated, non-redundant database of protein sequences with verified functional annotations. The training set contains 433,583 sequences and the validation set contains 22,821 sequences, with no overlap between them. Sequences longer than 512 tokens are truncated.

3.3 Training Configuration

Table 2 lists the training hyperparameters. The model is trained with the AdamW optimizer [10] using a linear warmup of 1,000 steps followed by cosine decay to near-zero learning rate. All training is performed on a single Cambricon MLU370 accelerator.

Table 2: Training Configuration

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Batch size	32 sequences
Sequence length	512 tokens
Warmup steps	1,000
Total steps	$\sim 13,500$ per epoch
Training epochs	5
Masking ratio	15%
Weight decay	0.01

4 Results

4.1 Training Dynamics

The model converges smoothly over 5 epochs. Figure 1 (left) shows the training and validation loss curves. The validation loss decreases from 0.4195 (epoch 1) to 0.4170 (final best), demonstrating stable training without overfitting.

Training / Val Loss (placeholder)	Perplexity over Training (placeholder)
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Figure 1: Training dynamics. Left: loss curves. Right: perplexity curves.

Table 3 reports the per-epoch training and validation metrics.

Table 3: Training Metrics per Epoch				
Epoch	Train Loss	Val Loss	Val PPL	
1	0.4195	0.4195	1.54	
2	0.4235	0.4218	1.55	
3	0.4219	0.4195	1.54	
4	0.4173	0.4185	1.54	
5	0.4185	0.4179	1.54	
Best (epoch 5, step 66000)	—	0.4170	1.54	

Training throughput is approximately 30,000 tokens/second per step, yielding roughly 2 hours per epoch and approximately 10 hours total for 5 epochs on a single MLU370 card.

4.2 Zero-Shot Mutation Prediction on GFP

We evaluate zero-shot mutation effect prediction using the Green Fluorescent Protein (GFP) benchmark, a standard reference in protein engineering [14]. Following the protocol of ESM-1v [12], we compute the difference in per-token log-likelihoods between wild-type and mutant sequences. The WT score is the average MLM logit at each masked position for the native sequence; the mutant score is computed analogously for the mutated sequence. The mutation effect Δ is the difference between mutant and WT scores.

Table 4 reports the zero-shot predictions for four GFP mutations with known functional annotations.

Table 4: Zero-Shot GFP Mutation Predictions				
Mutation	Type	WT Score	Mut Score	Δ
K7V	neutral	1.39	1.52	+0.13
K7I	neutral	1.39	0.40	−0.99
G66Y	brighter	1.70	0.33	−1.37
G66H	dimer	1.70	0.32	−1.38

The model correctly identifies K7V as a near-neutral mutation ($\Delta = +0.13$), distinguishes K7I as destabilizing ($\Delta = -0.99$), and captures the large deleterious effects of G66Y and G66H. Across these four cases, the Spearman rank correlation between predicted Δ and reported functional effect is $\rho = 0.200$. This modest but non-zero correlation is consistent with expectations for a small model evaluated zero-shot without task-specific adaptation, and establishes a baseline for future improvements.

5 Discussion and Limitations

ESM2-small demonstrates that the ESM-2 architecture achieves reasonable convergence on a compact 9.6M-parameter scale. The zero-shot GFP correlation of 0.200, while below state-of-the-art (ESM-2 150M achieves $\rho \approx 0.40$ on ProteinGym), confirms that evolutionary signals are partially captured even at small scale.

Several limitations should be noted. First, the model was trained for only 5 epochs; longer training or larger data (UniRef50 vs. Swiss-Prot) may improve performance. Second, the GFP benchmark is a small, curated set; a comprehensive evaluation on ProteinGym [1] would provide a more rigorous assessment. Third, the model lacks structural awareness; combining with AlphaFold2 embeddings or MSA information could improve functional predictions.

6 Conclusions

We present ESM2-small, a 9.6M-parameter protein language model trained on Swiss-Prot. The model achieves stable convergence to val loss 0.417 and demonstrates non-trivial zero-shot mutation prediction on GFP ($\rho = 0.200$). Full training code, data pipeline, and pre-trained weights are publicly available. Future work includes training on larger corpora, fine-tuning on specific protein families, and integration with structure prediction pipelines.

Reproducibility

- Code: <https://github.com/junior1p/ESM2-small>
- Model weights: <https://github.com/junior1p/ESM2-small/releases/tag/v1.0.0>
- Training data: Swiss-Prot (downloaded via rest.uniprot.org)

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